Speaking to Edit: LLM-Guided Semantic Interactive Part Editing of AI-generated 3D Models in Immersive Virtual Environments

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Abstract. We present a LLM-guided framework for speech-driven, part-level editing of AI-generated 3D models in immersive virtual reality. The system operates on intermediate 3D Gaussian representations and combines geometric clustering, category-specific semantic vocabularies, LLM-based command parsing, and structured Gaussian transformations. Unlike full model regeneration, the system applies local edits directly to labeled Gaussian primitives and provides closed-loop relabeling after modification. We implement the framework in a Unity-based VR prototype supporting controller-based manual interaction and label-semantic speech-to-text interaction. We present and evaluate an end-to-end system for speech-driven, semantic editing of AI-generated 3D models in VR. In a within-subjects study with 24 participants, we compare these two conditions across simple and multi-step editing tasks. Our Label-Semantic Speech-to-Text reduces completion time for Add and Modify tasks but is slower for simple Delete operations, while also receiving higher usability and satisfaction ratings. These results provide functional evidence that semantic speech can accelerate multi-step part editing in VR.

Keywords: LLM-guided semantic editing · Part-level model editing · Speech-driven interaction · Virtual reality editing

1 Introduction

Recent advances in generative models have significantly accelerated the creation of high-quality 3D assets, enabling a wide range of applications in immersive virtual environments, such as virtual reality (VR) content authoring, simulation, digital twins, and metaverse platforms. Despite their impressive visual fidelity, AI-generated 3D models remain difficult to edit and customize. Unlike traditional artist-created or CAD models that expose explicit part hierarchies and semantic structures, generative 3D models

are typically represented as holistic, unstructured entities, where component boundaries and semantic meanings are implicit. Consequently, even simple part-level modifications—such as resizing, repositioning, or replacing a specific component—often require re-running the entire generation pipeline, leading to high computational cost, long waiting times, and unpredictable results caused by ambiguous textual descriptions and limited controllability of generative processes.

In this paper, we introduce an LLM-guided semantic editing framework that enables intuitive, part-level manipulation of AI-generated 3D models directly inside immersive VR environments. Rather than relying on costly re-generation or brittle post-processing, our approach operates on intermediate 3D Gaussian representations produced by modern generative pipelines, turning unstructured outputs into semantically grounded, editable parts. A hybrid semantic labeling stage combines geometric clustering with in-context LLM reasoning to assign robust part labels to Gaussian primitives, and a text driven editing pipeline converts spoken or typed natural-language commands into structured, machine-executable part operations (e.g., scale, translate, rotate, stretch, delete). We implement this pipeline in a real-time Unity-based VR system with closed-loop relabeling to preserve global semantic consistency after edits, and validate the method through user studies that demonstrate improved efficiency and user satisfaction compared to manual and vanilla speech-to-text editing workflows.

This paper makes the following contributions:

- We proposed a semantic editing pipeline that binds category-specific part labels to Gaussian primitives using geometric features and LLM-based inference.
- We designed a structured language-to-edit interface that converts spoken commands into executable part-level transformations.
- We implement a VR prototype integrating speech input, semantic labeling, Gaussian editing, and closed-loop relabeling.

2 Related Work

2.1 3D Model Representation

3D model representation has long been a foundational topic in computer graphics and geometry processing. Traditional polygonal meshes remain the most widely used format due to their explicit surface topology, compatibility with physically based animation, and efficient real-time rasterization[1]. Point clouds and voxel grids, as topology-free alternatives, have also been extensively studied for simplifying geometric processing pipelines[33,29]. However, these representations often suffer from limited resolution or insufficient surface smoothness, making them less suitable for fine-grained editing.

Recent advances in neural scene representations—particularly Neural Radiance Fields (NeRF)[30] and 3D Gaussian Splatting (3DGS)[24]—have demonstrated strong capabilities in novel-view synthesis and high-fidelity reconstruction. 3DGS, in particular, provides an explicit and structured set of Gaussian primitives that support real-time rendering while retaining differentiable geometric attributes (e.g., scale, orientation, and opacity). These properties make Gaussian splatting especially suitable for interactive editing and semantic operations.

2.2 Structured Part–Hybrid Model Representations

Hybrid model representations commonly combine explicit methods to capture a model's coarse, interpretable structure (using semantically meaningful parameters) with implicit methods to encode fine geometric detail (using high-dimensional, hard-to-interpret codes). Local Deep Implicit Functions (LDIF) [14] are a representative example of this structured part–hybrid approach: LDIF represents object parts with a set of explicit Gaussian spheres while modeling surface detail with local implicit functions encoded by high-dimensional descriptors. This structured part–hybrid paradigm has inspired subsequent work—e.g., SPAGHETTI [19] and SALAD [25]—that build closely on LDIF. Notably, SALAD integrates diffusion models, enabling text-conditioned generation and substantially improving output quality, marking an important advance for 3D model synthesis.

There is also a rich body of work on 3D representations specifically tailored for editing. CNS-Edit [22] combines global latent variables with neural volumes to support point-specific edits finer than part-level manipulations. NeRF-based methods [30] and their variants [12] leverage multi-view imagery to reconstruct highly photorealistic 3D scenes and objects, but their native lack of explicit geometric structure makes direct semantic editing challenging [32,36]. Voxel-based approaches remain relevant, focusing on generation efficiency and final quality [31,11]. Systems like Trellis [38] introduce large pretrained models into 3D generation and are representation-agnostic—offering strong generalization across formats but generally not targeting editability. Other works exploit cross-domain data to aid 3D synthesis: Part123 [27] uses 2D segmentation to guide 3D part generation, while ANISE [8] assembles models from a preprocessed 3D part dataset. Collectively, these efforts contribute complementary strengths across generation and editing, with varying trade-offs between fidelity, controllability, and editability.

2.3 Deep Generative Networks and 3D Models

With the rapid development of deep learning, deep generative models have achieved remarkable results in image synthesis, data augmentation, and representation learning, and have increasingly been applied to the more challenging task of 3D content generation. Among these models, Generative Adversarial Networks (GANs) [16] and diffusion models [20] represent two influential and widely used paradigms.

GANs and their variants have been applied to 3D generation tasks—e.g., directly producing voxels or point clouds [37,15,26]. However, applying standard GAN architectures to 3D data with complex structure and variable topology often requires special design, motivating exploration of alternative generative frameworks [9]. Early theoretical work on diffusion processes laid the foundation for modern diffusion models [35], which recent advances such as DDPMs [20] and classifier-free guidance [21] have substantially developed. Diffusion models now achieve state-of-the-art results across modalities including images, audio, and 3D objects. Compared with GANs, diffusion models typically produce higher-fidelity and more diverse samples, better covering the data distribution and mitigating mode collapse [6]. Moreover, diffusion models naturally accommodate conditional generation: by providing conditioning signals (e.g.,

class labels, text embeddings, or images) to the denoising network, one can precisely control generated content [39]. This capability has driven their success in text-to-image synthesis and rapid extension into text-to-3D generation [23,25,27,5,28,40,7,3,4].

3 Method Overview

In this section, we present an overview of our proposed framework. Our approach operates on the LDIF representation [13,25], which provides an editable intermediate form for AI-generated 3D models. By leveraging large language models (LLMs) to extract structured part-level semantic labels, our system enables speech- and text-driven semantic editing directly on 3D Gaussian components, allowing users to intuitively modify model parts through natural language commands.

Our part-level model editing framework is built upon the structured AI-generated model representation provided by LDIF. As illustrated in Figure 1, the system consists of two main stages: model generation and model editing. In the model generation stage, users first provide spoken generation instructions, which are parsed and transformed into structured textual prompts using a large language model (LLM). These prompts are then fed into a Gaussian generation module, which employs a pretrained conditional diffusion model to iteratively denoise random noise or latent variables under the guidance of the encoded textual semantics, producing an initial set of 3D Gaussian parameters $\{g_i\}_{i=1}^{N}$. These generated Gaussian primitives not only serve as the foundation for subsequent fine-detail synthesis, but more importantly, explicitly correspond to editable model parts, making them directly accessible for interactive editing. Therefore, the Gaussian parameters are forwarded to the part editing module, allowing users to modify them interactively after generation. Subsequently, the refined Gaussian representation is passed through a high-dimensional implicit decoding module and a mesh reconstruction module to produce the final 3D geometry.

In the model editing stage, users directly issue spoken or textual editing commands. Similarly, these commands are first parsed and interpreted by the LLM. Meanwhile, we perform semantic binding on the Gaussian representation of the generated model, assigning hybrid semantic labels to each Gaussian primitive to establish explicit part-level semantic correspondence. Guided by the parsed editing intent, a text-driven part-level editing module modifies the parameters of the relevant Gaussian primitives to achieve targeted part editing. Finally, the edited Gaussian representation is decoded through the implicit and mesh reconstruction modules, yielding the updated 3D model. The hybrid semantic labeling module and the text-driven part-level editing module are detailed in the following sections.

4 Speech-Driven Model Part Editing

To enable speech-driven, part-level editing of 3D models, we propose a hybrid semantic annotation and text-driven editing framework (Figure 2). The framework combines geometric clustering with the commonsense reasoning of large language models to achieve training-free semantic part labeling—requiring no additional training or part-level annotations. We also develop a command parser that converts natural-language editing

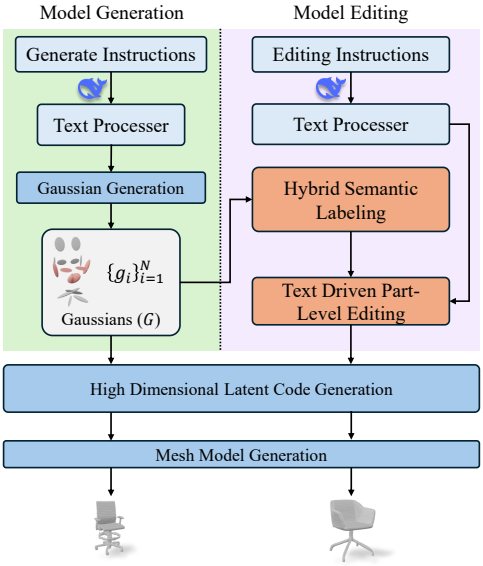


Fig. 1. Overview of our method for 3D model Part Editing.

requests into structured geometric operations, enabling fine-grained manipulation of AI-generated models represented by Gaussian primitives. Speech input is captured and transcribed in real time using Unity's built-in microphone API. The following subsections describe the framework's two core components in detail: the hybrid semantic labeling module and the text-driven part-level editing pipeline.

4.1 Hybrid Semantic Labeling

This module converts a 3D model's raw Gaussian representation into a semantically labeled Gaussian representation, where the label vocabulary is defined per object category (e.g., a chair may include {leg, seat, back, arm, etc.}). We formulate semantic labeling as an in-context learning problem for a large language model (LLM) via prompt engineering. The pipeline consists of two stages: feature extraction and LLM-based semantic inference, followed by domain-specific post-processing rules to enhance robustness.

Feature Extraction We first extract the spatial center of each Gaussian primitive and apply DBSCAN clustering [10]:

$$\text{cluster_ids} = \text{DBSCAN}(\{\mu_i\}_{i=1}^N, \varepsilon, \text{min_samples}) \tag{1}$$

The neighborhood radius ε is adaptively determined by the diagonal length d_{bbox} of the object's bounding box (Eq. 2), ensuring consistent scale across different objects:

$$\varepsilon = \max(0.03 \times d_{\text{bbox}}, 0.015) \tag{2}$$

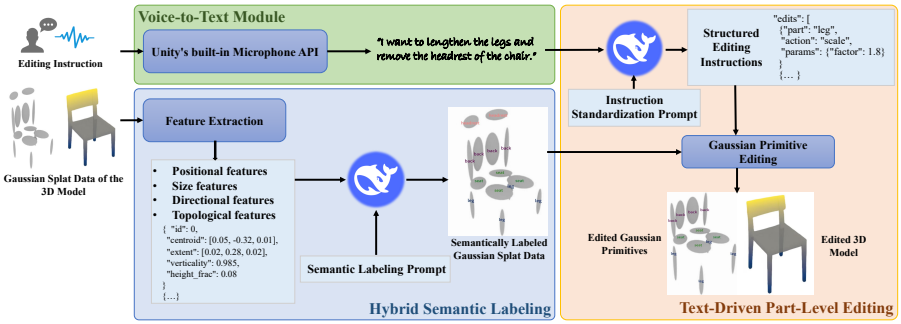


Fig. 2. Our proposed Speech-Driven Model Part Editing System consists of three core modules: (1) Voice-to-Text Module: Converts user voice commands into natural language text. (2) Hybrid Semantic Labeling: Extracts geometric features from the Gaussian splatting representation of the 3D model using our Feature Extraction Algorithm, and inputs them into an LLM to produce semantically labeled Gaussian data. (3) Text-Driven Part-Level Editing: Utilizes the LLM to parse natural language instructions into structured editing instructions, which are then executed through our Gaussian splat editing algorithms to modify the 3D model accordingly.

This strategy lets the clustering scale adapt to object size and prevents fixed thresholds from failing in diverse scenarios. Noise points are reassigned as singleton clusters so that every Gaussian primitive has a clearly defined cluster membership.

LLM-based Semantic Inference Large language models (LLMs) have shown strong capabilities in natural language understanding and generation, but applying them to 3D geometric reasoning poses several unique challenges: (1) LLMs lack direct perception of 3D spatial structure; (2) the mapping from geometric features (e.g., verticality, height fraction) to semantic concepts (e.g., "leg", "seat") is implicit and domain dependent; and (3) the stability and parsability of the output format directly affect system robustness. To address these issues, we design a systematic prompt-engineering framework that combines three key strategies: structured input representation, few-shot example guidance, and constrained output formatting. This enables the LLM to perform training free geometric-to-semantic mapping and robust instruction parsing, providing a reliable semantic basis for speech-driven, part-level editing. The system prompt follows a three-stage structure that defines the task, injects domain knowledge, and constrains the output format.

4.2 Text-Driven Part-Level Editing Pipeline

This module converts natural-language editing commands (e.g., "make the chair legs thicker" or "remove the armrests") into structured instructions for part-level editing of 3DGS models. We use an LLM to parse the input text and generate a structured editing plan. Each operation is represented as a triplet (p, a, θ), where p is the target part, a is the action type ($a \in A$, $A = \{\text{scale}, \text{stretch}, \text{rotate}, \text{translate}, \text{delete}, \dots\}$), and θ denotes the operation parameters.

We design a structured system prompt to ensure that the LLM reliably maps natural-language commands to machine-executable operations. The prompt specifies the task definition, the allowed operations, and strict output-format constraints. To prevent hallucinated actions and unstable output structures, we explicitly define the semantic label set and action space.

Given the parsed editing plan, the system applies the corresponding geometric transformations in order:

- Scale: isotropic scaling around the part centroid;
- Stretch: anisotropic scaling along a specified axis while preserving local distribution;
- Rotate: rotation using the Rodrigues formula to ensure valid $\mathrm{SO}(3)$ transformations;
- Delete: opacity adjustment for soft deletion, enabling subsequent undo operations.

We further implement a closed-loop feedback mechanism: after editing, the updated Gaussian geometry is reprocessed by the hybrid semantic-labeling module to refine part assignments and maintain global semantic-topology consistency. The proposed text-driven part-level editing pipeline supports a broad range of affine and non-linear transformations, offering strong generalization and interpretability. Without retraining the LLM for new object categories, our method can directly produce high-quality, part-level semantic edits.

Figure 3 illustrates our editing results on a generated chair and table. Before editing, both models are represented as semantically labeled Gaussian primitives, allowing users to directly manipulate specific Gaussians based on their semantic tags. In the top example, the user issues the command "Rotate the chair legs outwards by 30 degrees," which drives edits on the corresponding Gaussian primitives and results in a rotated leg configuration in the 3D chair model. In the bottom example, the command "Raise the table drawer a little" adjusts the relative height of the drawer in the table model. The final edited geometry preserves a plausible table shape by slightly compressing the drawer's thickness while lifting it upward.

4.3 Interactive Editing in a VR Environment

We implement a VR editing system that supports both speech-based and manual interaction for real-time manipulation of 3D models represented by Gaussian primitives. Users can edit models either by issuing spoken commands or by directly manipulating Gaussian proxies with VR controllers. The frontend is built in Unity and is responsible for capturing controller input, handling microphone-based speech input, and providing immediate visual feedback. It serves as the interaction layer where users adjust the position and orientation of Gaussian spheres, as well as the pose of individual object components in the virtual room. The backend, implemented in Python, handles computationally intensive tasks such as deep learning inference, complex geometric computation, and Gaussian-based model generation.

Users interact with the model via the PICO controllers' thumbsticks and buttons. As illustrated in Figure 4, they can control both the object and its Gaussian spheres in the

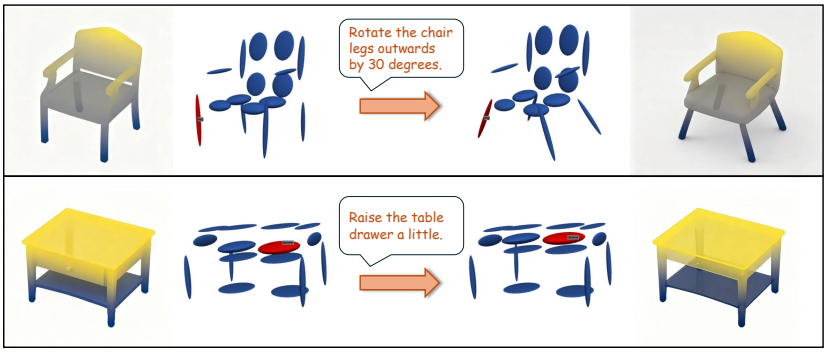


Fig. 3. Two example edits using LS-STT. Top: the user adjusts the orientation of the chair legs via speech commands, producing a plausible edited chair model. Bottom: the user raises the drawer of the table through a spoken command, resulting in a reasonable edited table geometry.

VR scene: moving the thumbstick up and down translates the object along the x-axis (forward/backward), moving it left and right rotates the object horizontally; moving the controller itself translates the object along the y- and z-axes; rotating the controller performs vertical rotations.

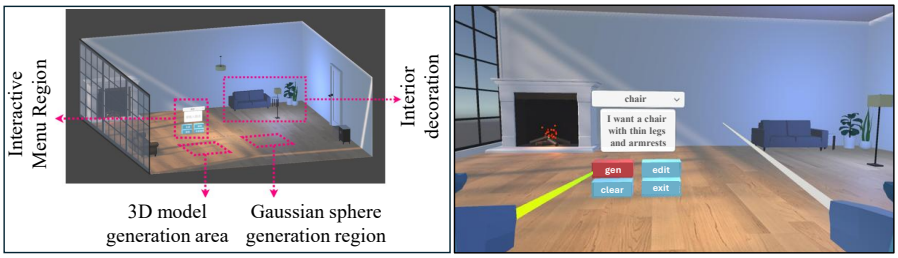


Fig. 4. Interactive 3D model editing scene in VR. Left shows the layout-view of the scene. There are mainly four regions: interactive menu region, 3D model generation region, gaussian sphere generation region and interior decoration region. Right shows the first view of a participant.

The overall interaction workflow is as follows. At the beginning of each trial, the user wears the HMD and enters a virtual living room, then sequentially selects an object category (chair, table, or airplane from our dataset), specifies textual constraints, and provides functional instructions. Once the object and its associated Gaussian spheres are displayed in the UI, the user can fine-tune the spheres to further edit the object's 3D shape. The backend then processes these edits and generates the updated result. Please refer to the accompanying video for additional details.

5 User Study

5.1 Overview and Hypothesis

To evaluate the effectiveness of our label-semantic speech-to-text editing method for 3D model manipulation in virtual reality (VR), we conducted a formal within-subjects user study. The study empirically compares two editing approaches: Manual Interaction (MI) and Label-Semantic Speech-to-Text (LS-STT). The primary aim is to determine whether LS-STT—by combining semantic object labels with spoken commands—can improve editing efficiency and user satisfaction.

5.2 Participants and Apparatus

We recruited 24 participants, with a gender distribution as balanced as possible. Participants were between 19 and 25 years old. All participants reported normal vision or corrected-to-normal vision. None of them had participated in any prior pilot studies related to this project.

The experiment was conducted using a Pico 4 Pro head-mounted display (HMD) equipped with handheld controllers. The VR system was driven by a workstation with an Intel i9 CPU, 16GB of RAM, and an NVIDIA GeForce RTX 4090 graphics card. The experiment was carried out in a tracked physical space of approximately $4\,\mathrm{m} \times 4\,\mathrm{m}$.

The VR editing system was implemented using the Unity engine and supported real-time manual interaction, speech-based text input, and semantic-tag-based command parsing. Before each experimental session, the inter-pupillary distance (IPD) of each participant was measured and adjusted to ensure optimal visual comfort. The virtual environment was rendered at a stable frame rate suitable for immersive interaction.

5.3 User Study Design

We conducted a repeated-measures, within-subjects user study to evaluate the two 3D model editing methods: manual interactive editing (MI) and label-semantic–enhanced speech-to-text editing (LS-STT). Each participant completed the editing tasks using both methods.

For each method, participants performed a set of model-editing tasks comprising add, delete, and modify operations. Modify tasks required translating, rotating, and scaling objects. After finishing individual tasks, participants completed an integrated editing task that combined multiple operations. Before each method, participants were given sufficient time to familiarize themselves with the corresponding interaction technique. Once ready, they performed the tasks independently while objective performance metrics were recorded.

After completing all tasks for a given method, participants filled out a post-condition questionnaire including the NASA Task Load Index (NASA-TLX) [18,17], the System Usability Scale (SUS) [2], and a subjective task satisfaction rating on a 7-point Likert scale. This procedure was repeated until both methods had been tested. The independent variable was editing method. Dependent variables included task completion time and subjective user feedback.

5.4 Procedure and Metrics

Upon arrival, each participant signed an informed consent form and completed a demographic questionnaire to establish baseline data. An experimenter then explained the VR setup and the 3D model editing procedures. The study used a repeated-measures design: every participant performed the task set with all methods (MI and LS-STT). To mitigate learning and fatigue effects, method order was counter balanced using a Latin square.

The procedure for each method was identical. In the training phase, participants received a standardized tutorial and practiced the specific interaction techniques (e.g., controller operations for MI, voice-command syntax for LS-STT) until they felt proficient and ready. In the task phase, participants completed three tasks in a fixed order: Add, Delete and Modify (an Integrated task combining translating, Rotating, and scaling). Each task was performed individually and objective metrics were recorded. After completing all tasks for a given method, participants removed the head-mounted display (HMD) and completed post-condition questionnaires: NASA-TLX and SUS. They also rated satisfaction for each individual task and for the integrated task on a 7-point Likert scale (1 = very dissatisfied, 7 = very satisfied). Participants were given at least two minutes of rest between methods and proceeded to the next condition when ready. Each session lasted approximately 30 minutes. We recorded the task completion time t (total interaction time, in seconds, to complete each individual and comprehensive editing task) for both MI and LS-STT methods.

5.5 Results and Discussion

We analyze the values of our metrics using the p value of a t-test [41], and we quantify effect size using Cohen's d [34]. Table 1 gives the task completion time results for the three editing tasks of the two methods. For the Add task, our LS-STT method reduced mean completion time by 51.1% compared to the baselines. This difference is statistically significant ($p < 0.001$) and corresponds to a very large effect size. For the Delete task, in contrast, LS-STT increased mean completion time by 41.8%; this difference is also statistically significant ($p < 0.001$) with a very large effect size. For the Modify task, LS-STT reduced mean completion time by 72.0%, again with a statistically significant difference ($p < 0.001$) and a very large effect size.

Table 1. Task completion time t, in seconds

Task	Method	Avg± std.	Red.	p	Cohen's d	Effect size
Add	MI	17.54 ± 2.89	51.1%	$< 0.001^*$	4.01	Huge
	LS-STT	8.58 ± 1.29				
Delete	MI	5.08 ± 1.26	-41.8%	$< 0.001^*$	1.57	Very Large
	LS-STT	7.21 ± 1.44				
Modify	MI	162.54 ± 15.15	72.0%	$< 0.001^*$	9.79	Huge
	LS-STT	45.5 ± 7.48				

For the Delete task, LS-STT actually led to longer editing times. We attribute this to the simplicity of the deletion operation, which in the manual condition only requires selecting a Gaussian sphere and pressing a delete button. Speech-based input, however, necessarily incurs a minimum utterance and recognition time, making manual editing more efficient for such simple, single-step actions. In contrast, for the more complex Add and Modify tasks, LS-STT clearly outperforms manual interaction, substantially improving participants' editing efficiency. Overall, these results indicate that the proposed LS-STT method can effectively reduce the time required for part-level 3D model editing, particularly for tasks involving more complex or multi-step operations.

Figure 5 presents the NASA-TLX and SUS scores for the two methods. We conducted Holm–Bonferroni–corrected post-hoc paired t-tests to compare subscale scores between MI and LS-STT. For both the overall NASA-TLX score and the SUS score, LS-STT significantly outperformed MI ($p < 0.001$). For the NASA-TLX subscales, there was no significant difference between the two methods in Mental Demand and Performance ($p = 0.459$, $p = 0.912$). For the remaining four subscales, LS-STT showed statistically significant improvements over MI.

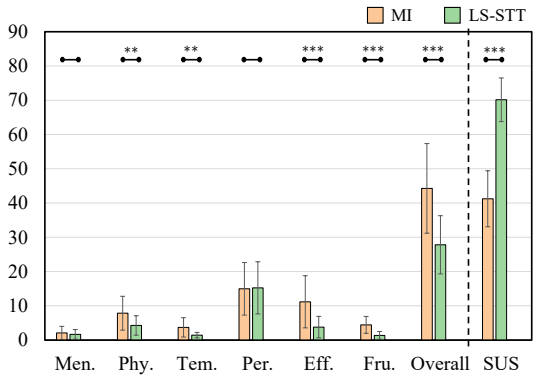


Fig. 5. Plots for NASA-TLX subscores (left of the vertical dash line) and SUS scores (rightmost column) of the compared two methods. Statistically significant differences are marked with asterisks.

As shown in Table 2, we report the mean Likert scores for MI and LS-STT and use p-values and Cohen's d to assess differences between the two methods. The results indicate a significant advantage of our LS-STT method over MI, with LS-STT being consistently preferred by participants. In post-experiment informal interviews, participants reported that LS-STT was more enjoyable to use, as they could modify 3D models simply by verbally describing the desired changes. However, some participants also noted occasional mismatches between the executed edits and their precise intentions. Overall, LS-STT was perceived as more engaging and preferable than the traditional MI approach.

Table 2. Likert scale comparison between our MI and LS-STT methods.

Method	Avg± std.	Red.	p	Cohen's d	Effect size
MI	3.88 ± 0.73	22.6%	0.003^*	0.92	Large
LS-STT	4.75 ± 1.13				

6 Conclusion

We presented an LLM-guided framework for semantic, part-level editing of AI-generated 3D models in immersive VR environments. Instead of relying on expensive re-generation or post-hoc mesh processing, our approach operates directly on intermediate 3D Gaussian representations, combining hybrid semantic labeling with a text-driven editing pipeline. By integrating geometric clustering with in-context LLM reasoning, we obtain training free part-level semantic bindings for Gaussian primitives, and we leverage constrained prompt engineering to translate natural-language commands into structured, machine-executable editing operations. We implemented a real-time Unity-based VR system that supports both manual and speech-based interaction, and introduced a closed-loop relabeling mechanism to maintain semantic consistency after edits.

Our within-subjects user study shows that the proposed label-semantic speech-to-text (LS-STT) method significantly improves task completion time for more complex Add and Modify tasks and yields higher usability and satisfaction scores than traditional manual interaction, while being slightly less efficient for very simple Delete operations. Overall, participants preferred LS-STT and found it more engaging and expressive for part-level 3D editing.

Our semantic-labeling module relies partly on geometric assumptions encoded through DBSCAN features, object-relative position, verticality, and category-specific correction rules. These assumptions are effective for the upright furniture models evaluated in this study, but they may fail for rotated, asymmetric, articulated, or non-furniture objects. Future work should replace these rules with object-centric coordinate normalization, learned or rotation-invariant geometric descriptors, uncertainty-aware labeling, and evaluations across substantially broader asset categories.

Acknowledgments. This work is supported by the New Generation Artificial Intelligence-National Science and Technology Major Project 2025ZD0124004, the National Natural Science Foundation of China through Projects 62507002 and 62372026, and the Fundamental Research Funds for the Central Universities.

Disclosure of Interests. The authors have no competing interests to declare that are relevant to the content of this article.

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